CHEMISTRY Department
SIMHADRI

Note No :1

Date: 13/10/2025(19:53:04)

Ref:SMTPP/Chemistry/2025-26/SIMCHEM/449840

Subject:LMI REVISION FOR Chemical Control of the Water-Steam Circuits of Drum Once Through Type Boilers and HRSG.

Unauthorized Use Prohibited

Previous LMI Version: LMI/OD/OPS/CHEM/001 Rev. No.: 10

Present LMI Vesrion: LMI/OD/OPS/CHEM/001, Rev. No.: 11

Changes in LMI are briefed in the below.

1. ‘ACC’ (After Cation Conductivity) has been replaced with ‘CC’ (Cation Conductivity) in all parameter tables, as the abbreviation ‘ACC’ is used for Air-Cooled Condenser in this TC document.

2.Updated Sec. 5.3 (Note-a & b) • Incorporated guidelines for the auto operation of LP dosing pumps (ammonia and hydrazine) and oxygen dosing system. • Specified additional ammonia dosing points to maintain the pH of the steam-water circuit, as and when required.

3.Updated Sec. 5.5 (i) Incorporated guidelines for the auto operation of HP dosing pumps.

4.The DM makeup CC analyser has been removed from the core parameters list for all unit types.

5.Updated Clause 7.2 Incorporated low-level chloride under diagnostic monitoring parameters

6. Updated Clause 8.1 Table-2 Incorporated monitoring of TOC in condensate water samples along with recommended limits.

7. Updated Clause 8.3 Table-4 Incorporated feedwater specific conductivity limits for all types of units. Feed-water DO limit has been revised for both mixed metallurgy and all-ferrous units.

8. Updated Clause 8.4 Table-5. Revised the specific conductivity limit for drum-type units. Included monitoring of sodium in boiler water samples, with limits for both phosphate and caustic treatment regimes.

9. Updated Annexure-I Incorporated monitoring of sodium.

Draft LMI and OD is attached for reference.

Submitted for approval please.

Submitted by:

आर. बालकृष्ण पट्नायक - उप प्रबंधक (रसायन)

R Balakrishna Patnaik - 103088 (DEPUTY MANAGER(CHEMIST) ,CHEMISTRY) ,Simhadri (RBALAKRISHNAPATNAIK@NTPC.CO.IN, 9437494990)
On:13/10/2025(19:53:04) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:2

Submitted for comments of concerned departments and approval by CA please.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY) (SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri
On:14/10/2025(09:32:06) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:3

Please review.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG) (NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:14/10/2025(10:44:04) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited

Unauthorized Use Prohibited

Note No:4

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:14/10/2025(18:22:37) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:5

Modified accordingly and attached.

Submitted for approval please.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri
On:15/10/2025(11:16:11) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited
Unauthorized Use Prohibited

Note No:6

Clause no. 11 to be revised in line with Simhadri station LMI format

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:15/10/2025(12:09:05) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:7

Document is revised and attached. May be approved.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri
On:16/10/2025(15:48:25) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited
Unauthorized Use Prohibited

Note No:8

Reviewed

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG) (AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:17/10/2025(17:50:36) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:9

LMI is in line with the OD. Forwarded for further review.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG) (NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:18/10/2025(10:23:06) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited

Note No:10

Pl examine.

Submitted by:

सुप्रितिम कुरी - उप महाप्रबंधक

SUPRATIM KURI - 100946 (DY. GENERAL MANAGER ,MECH MAINT- BOILER) (SUPRATIM@NTPC.CO.IN, 9650998020) ,Simhadri
On:18/10/2025(20:12:24) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:11

Forwarded for further review please.

Submitted by:

शशि कांत - उप महाप्रबंधक

SHASHI KANT - 101559 (DY. GENERAL MANAGER ,MECH MAINT- BOILER) (SHASHIKANT@NTPC.CO.IN, 9425178426) ,Simhadri
On:21/10/2025(19:48:42) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:12

Submitted by:

सुप्रितिम कुरी - उप महाप्रबंधक

SUPRATIM KURI - 100946 (DY. GENERAL MANAGER ,MECH MAINT- BOILER) (SUPRATIM@NTPC.CO.IN, 9650998020) ,Simhadri
On:21/10/2025(20:24:31) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited

Note No:13

Please review

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE) (NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri
On:22/10/2025(09:35:17) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:14

LMI reviewed and is in line with the revised OD.
Forwarded for approval.

Submitted by:

के पी रमेश - उप महाप्रबंधक

KANADE PIYUSH RAMESH - 101897 (DY. GENERAL MANAGER ,MECH MAINT- TURBINE) (PIYUSHKANADE@NTPC.CO.IN, 9425219272) ,Simhadri
On:22/10/2025(16:08:27) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:15

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE) (NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri
On:22/10/2025(16:19:09) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited

Note No:16

Checked and forwarded.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri
On:28/10/2025(17:06:51) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:17

LMI reviewed.

Submitted by:

अनिल कुमार मौर्या - अपर महाप्रबंधक

Anil Kumar Maurya - 008631 (ADDL. GENERAL MANAGER ,C & I MAINT)
(ANILKUMARMAURYA@NTPC.CO.IN, 8004940574) ,Simhadri
On:29/10/2025(09:34:19) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited
Unauthorized Use Prohibited

Note No:18

LMI is in line with the OD. May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:29/10/2025(11:52:44) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:19

Submitted by:

हेमेन्द्र कुमार वर्मा - महाप्रबंधक

Hemendra Kumar Verma - 006459 (GENERAL MANAGER ,MAINTENANCE)
(HKVERMA02@NTPC.CO.IN, 7091092468) ,Simhadri
On:31/10/2025(11:00:19) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Note No:20

Submitted by:

अहमद कौनैन रज़ा - महाप्रबंधक

Ahmad Kaunain Raza - 006151 (GENERAL MANAGER ,O & M)
(AKRAZA@NTPC.CO.IN, 9407064096) ,Simhadri
On:31/10/2025(17:17:02) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

Unauthorized Use Prohibited

Note No:21

Unauthorized Use Prohibited

Approved & Submitted by:

समीर शर्मा - कार्यकारी निदेशक

Sameer Sharma - 005041 (EXECUTIVE DIRECTOR ,HEAD OF PROJECT)
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri
On:31/10/2025(17:37:40) **Ref:**SMTPP/Chemistry/2025-26/SIMCHEM/449840

NTPC **SIMHADRI**

LOCATION MANAGEMENT INSTRUCTION

Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG

LMI/OD/OPS/CHEM/001 Rev. No.: 11

Oct-2025

Approved for Implementation by: HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
2	LMI No.	LMI/OD/OPS/CHEM/001 Rev. No.: 11
3	Reference Document	COS-ISO-00-OGN/OPS/CHEM/019 Rev. No.: 02 July 2025
4	SupersededDocument	LMI/OD/OPS/CHEM/001 Rev. No.: 10 Aug-2023
5	Prepared By	Name: R. Balakrishna Patnaik, Dy. Manager (Chem)
6	Checked By	Name: Santanu Jana, D.G.M (Chem)
7	Reviewed By	Name: A.K Raza GM (O&M)
8	Approved By	Name: Sameer Sharma HOP (Simhadri)

INDEX

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

1.0 Introduction

The selection, optimization, and maintenance of cycle chemistry treatment regimes are essential for the efficient operation of power plants. Proper cycle chemistry ensures high availability, efficiency, and long-term reliability of power stations. If cycle chemistry is not carefully controlled and monitored within predefined limits, corrosion, deposition, and other forms of metal damage may occur throughout the plant system. Most cycle chemistry-related degradation phenomena are long-term in nature. Therefore, effective management of steam water cycle chemistry parameters is a critical factor in maintaining the health and longevity of boilers and turbines. Excellent cycle chemistry control ensures that cycle components reach their design life without excessive forced outages caused by corrosion, deposition, and scaling. It also prevents deterioration of cycle efficiency and plant generating capacity due to poor water chemistry. The practice of cycle chemistry control involves purification of makeup water, removal of impurities within the cycle (through blowdown, air removal, condensate polishing, and chemical cleaning, etc.), addition of water treatment chemicals, and continuous monitoring. Proper design, operation, and maintenance are also fundamental aspects of effective cycle chemistry management.

This document provides the recommended target values for key chemistry parameters at various points in the steam-water cycle that should be maintained during normal operation of power plants.

2.0 Superseded Documents

"Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG" No- LMI/OD/OPS/CHEM/001, Rev. No.: 10.

3.0 SCOPE

The LMI prescribes the various parameters and the limits for the chemical control of the water/steam circuit during steady state operation. The targets are essentially fixed for steady state operation of plant and the targets are to be achieved and maintained for at least 95% of the daily running. General aspects, Objectives, principles and Practices are briefly described.

Instrumentation, Standard analytical techniques, Frequency of manual testing and recording of online analyzers' readings are illustrated.

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

4.0 RECORDS OF REVISION

The following major changes have been incorporated from the previous version, as outlined below:

Sl. No	Rev.11
1	‘ACC’ (After Cation Conductivity) has been replaced with ‘CC’ (Cation Conductivity) in all parameter tables, as the abbreviation ‘ACC’ is used for Air-Cooled Condenser in this TC document.
2	Updated Sec. 5.3 (Note-a & b) • Incorporated guidelines for the auto operation of LP dosing pumps (ammonia and hydrazine) and oxygen dosing system. • Specified additional ammonia dosing points to maintain the pH of the steam-water circuit, as and when required.
3	Updated Sec. 5.5 (i) Incorporated guidelines for the auto operation of HP dosing pumps.
4	The DM makeup CC analyser has been removed from the core parameters list for all unit types.
5	Updated Clause 7.2 Incorporated low-level chloride under diagnostic monitoring parameters
6	Updated Clause 8.1 Table-2 Incorporated monitoring of TOC in condensate water samples along with recommended limits.
7	Updated Clause 8.3 Table-4 • Incorporated feedwater specific conductivity limits for all types of units • Feed-water DO limit has been revised for both mixed metallurgy and all-ferrous units.
8	Updated Clause 8.4 Table-5 Revised the specific conductivity limit for drum-type units. Included monitoring of sodium in boiler water samples, with limits for both phosphate and caustic treatment regimes.
9	Updated Annexure-I Incorporated monitoring of sodium.

5.0 OBJECTIVES OF WATER CHEMISTRY CONTROL

i. Prevention of corrosion in boiler, steam, condensate, feed water and air-cooled condenser (ACC) systems.
ii. Prevention of accumulation of scale and deposit formation on heat transfer surfaces.
iii. Reducing corrodent concentrations. iv. Maintenance of high level of steam purity to avoid deposits in turbines.

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

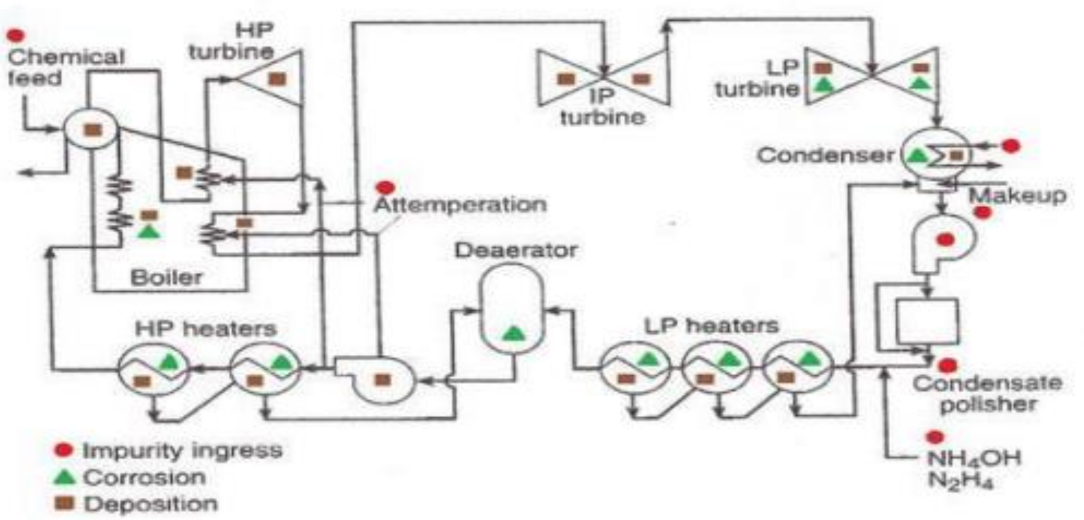
5.1 CYCLE CHEMISTRY GOALS

The following goals can be achieved through proper cycle chemistry control and monitoring, along with effective operation and maintenance:

i. No boiler tube failures related to cycle chemistry.
ii. No steam turbine problems (such as corrosion or deposits) related to cycle chemistry.
iii. No single-phase or two-phase flow-accelerated corrosion (FAC).
iv. Optimized cycle chemistry, boiler, and feedwater treatment for each unit.
v. Elimination of the need for chemical cleaning of boilers in both all-ferrous and mixed

The Scope of LMI is for four number of units with Drum type boilers at NTPC Simhadri for their start up, shutdown and Layup practices during shutdown of units.

5.2 CYCLE CHEMISTRY DIAGRAM



5.3 FEED WATER TREATMENTS

The three recommended feed water treatments are:

i. ALL VOLATILE TREATMENT (REDUCING), AVT(R) In this treatment, ammonia and hydrazine (reducing agent) are dosed at CEP discharge/ CPU outlet. Ammonia and reducing agents should be fed from separate metering tanks to ensure controlled dosing of each chemical. This feedwater treatment is used for mixed metallurgy units only. List of mixed metallurgy thermal units is enclosed as Annexure-III.
ii. ALL VOLATILE TREATMENT (OXIDIZING), AVT(O) In this treatment, no reducing agent (e.g. hydrazine) is added. Ammonia is dosed at CEP discharge/ CPU outlet. This feedwater treatment is applicable to all-ferrous units.
iii. OXYGENATED TREATMENT, (OT) Ammonia and oxygen are dosed at CPU outlet while oxygen is additionally dosed at the deaerator outlet, if required. This feedwater treatment is applicable to all supercritical units.

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

In NTPC Simhadri all four units are running on feed water treatment under AVT(O) regime.

Note: a) All the above-mentioned chemical dosing (ammonia, hydrazine, and oxygen) should be automated to always keep parameters within recommended limits, especially during unit load changes or other operational variations, to prevent corrosion in steam-water circuits. COS approved logic should be implemented for auto operation of ammonia dosing system.
b) Provision should be available for ammonia dosing at the suction of the boiler feed pumps and at the deaerator filling line from the boiler fill pump, to allow additional dosing as and when required

a) It is also essential that the core monitoring instruments (refer to 'COS-ISO-00-OD/OPS/CHEM/001' for list of analysers) be in service to provide reliable data as early as possible during the unit startup process.
b) The SWAS chiller with a standby compressor unit should be available.

1. The Express Lab should be available in the SWAS room for analyzing all chemistry parameters required during the startup and normal operation of the unit.
2. An online total iron analyzer should be used to monitor total iron in the feed water sample during the startup of a supercritical unit.
3. The startup cleanup time depends on the draining rate of open flushing and the circulation rate through the CPU for closed-loop flushing processes.
4. The condensate dumping arrangement, as specified in Annexure-11, should be available for draining high CRUD/TDS water into the atmosphere during startup. If condensate dumping line to atmosphere is not available, it should be provided during OH /Opportunity shutdown.

5.4. BOILER WATER TREATMENTS

There are essentially two types of boiler water treatment:

i. PHOSPHATE TREATMENT, (PT) Trisodium phosphate (Na_3PO_4) is added to the boiler drum from HP dosing pump to provide solid alkali-based pH control. The boiler chemistry control strategy focuses on maintaining an adequate concentration of solid alkali-either trisodium phosphate (TSP) or caustic (NaOH) in the drum to neutralize acidic contaminant ingress, such as chlorides, sulfates, etc. A minimum of 100 ppb sodium is always to be maintained in the boiler water. At the same time, it is critical to avoid excessive concentrations of caustic or TSP, as this can increase the risk of carryover to the turbine and lead to caustic gouging in boiler water wall tubes. In this treatment, sodium levels in Saturated steam (SS) and Main steam (MS) samples should be continuously monitored and maintained below 2 ppb to prevent carryover and ensure the protection of boiler and turbine components. HP dosing should be automated to prevent overfeeding of chemicals into the boiler drum. COS approved logic should be used for this purpose.

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

ii. CAUSTIC TREATMENT, (CT) Caustic is added to the boiler drum from HP dosing pump to provide solid alkali-based pH control. The boiler chemistry control strategy aims to maintain an adequate concentration of caustic (NaOH) in the drum to neutralize acidic contaminant ingress (such as chlorides, sulfates, etc.) while avoiding excessive levels. Overdosing of caustic can increase the risk of caustic gouging in boiler water wall tubes and carryover to the turbine steam path. However, a minimum of 100 ppb sodium is always to be maintained in the boiler water. In this treatment, sodium levels in both saturated steam and main steam samples should be continuously monitored and maintained below 2 ppb to protect turbine components. For units experiencing phosphate hideout, controlled dosing of caustic may be carried out at drum pressures below 160 kg/cm^2 to maintain boiler water alkalinity. In such cases, close monitoring of sodium levels (limit 2 ppb, max) in saturated and main steam samples is essential to minimize the risk of carryover.

In NTPC Simhadri all four units are running on Caustic treatment regime.

6.0 CYCLE CHEMISTRY MONITORING PARAMETERS AND THEIR IMPORTANCE

6.1 CONDUCTIVITY:
There are three types of conductivity measurements:
i. Specific conductivity (SC)
ii. Cation conductivity (CC)
iii. Degassed Cation Conductivity (DCC)

i. SPECIFIC CONDUCTIVITY (SC) is a reliable online method for monitoring the overall level of contamination, treatment chemicals and their trends.

ii. CATION CONDUCTIVITY (CC) is a critical monitoring parameter in the steam-water cycle, used to indirectly assess levels of anionic impurities (such as chloride, sulfate, organic acid, etc.) and provide early indication of contaminant in-leakage.

iii. DEGASSED CATION CONDUCTIVITY (DCC) is a variation of cation conductivity measurement. When applied correctly, it removes most of the CO_2 from the sample, allowing accurate detection of other anions, including organic acids. All three types of conductivity are critical core monitoring parameters, and their values are continuously monitored for one or more of the following reasons:

a) To maintain conductivity levels within acceptable limits. To facilitate the correlation of a water chemistry parameter (e.g., pH, conductivity, & ammonia correlation).
b) To check the accuracy of water chemistry control (such as ammonia or pH).
c) To warn of condenser tube leakage/ seepage.
d) To warn of condensate polisher malfunction.
e) To warn of in-leakage of contaminants.
f) To provide feedback signal for automated process control.

g) To facilitate demineralizer system operation and/or regeneration.

h) To monitor the intrusion of volatile contaminants (e.g., CO_2 or volatile organics) Note: Specific conductivity, cation conductivity, or degassed cation conductivity measurements are affected by the temperature of sample being measured. By convention, conductivity is routinely reported corrected to 25^0C. Even minor temperature differences from 25^0C can have a significant error on observed conductivity measurements.

6.2 pH :

pH is a critical core monitoring parameter. It should be continuously monitored online to check the acceptability of the water chemistry, thereby ensuring that corrosion rates are kept at low levels. In steam-water cycle pH is monitored for one or more of the following reasons:

a) To maintain pH levels within acceptable limits.

b) Corrosion of metals and alloys is a function of pH.

c) Alkaline pH values increase the stability of the oxide film and reduce oxide solubility in water.

d) To facilitate the correlation between two or more water chemistry parameters (e.g., pH, conductivity, ammonia correlation)

e) To provide a feedback signal for automated chemical dosing and process control.

f) To warn of condensate polisher malfunction.

g) To warn of in-leakage of contaminants.

Note: pH measurement is also affected by the temperature of the sample being measured. By convention, pH is routinely reported as corrected to 25^0C. Even minor temperature deviations from 25^0C can cause significant errors in observed pH measurements.

6.3 SODIUM (Na):

Sodium is a critical core monitoring parameter. It should be monitored continuously online to check the acceptability of water chemistry, thereby ensuring that corrosion rates are kept at low levels. Thus, sodium is monitored for one or more of the following reasons:

a) To maintain sodium levels within acceptable limits.

b) To warn of in-leakage of contaminants.

c) To warn of boiler water carryover.

d) To identify cooling water in-leakage at the main steam condenser and to warn of CPU malfunction.

6.4 DISSOLVED OXYGEN (DO):

Dissolved oxygen (DO) is a critical core monitoring parameter and is to be continuously monitored. Monitoring dissolved oxygen is necessary for the following reasons:

a) The oxygen level at the condensate pump discharge is an indication of air in-leakage into the cycle.

b) Single-phase flow accelerated corrosion of carbon steel components is eliminated by adding oxygen to high purity feed water.

c) Units with all-ferrous feed water systems operating on AVT(O) and OT, can reduce Iron transport by controlling dissolved oxygen in feed water systems.

6.5 SILICA (SiO2):

Monitoring of silica is necessary for the following reasons:

a) Precipitation of silica forms silicate deposits on the turbine that are not soluble in water and are very difficult to remove.

b) Silicate deposits cause losses in turbine capacity and efficiency and under extreme conditions.

c) Significant amounts of silica may enter the condensate/feedwater in a non-reactive colloidal form through the makeup, remaining undetected and causing boiler water and steam target values to be exceeded despite otherwise good makeup quality.

6.6 OXIDATION-REDUCTION POTENTIAL (ORP):

Monitoring of ORP is necessary for the following reasons:

a) The corrosion of copper alloys, and carbon steel are functions of the system ORP.

b) The maintenance of reducing conditions < –150 mV can effectively minimize the corrosion rate of copper alloys in power plant cycles.

6.7 CHLORIDE (Cl):

Monitoring of chloride for boiler drum water and steam is required because:

a) Chloride contributes to the following: Corrosion fatigue, Stress corrosion cracking (SCC), pitting in LP turbines, etc.

b) Chloride contributes to corrosion, hydrogen damage, and can cause pitting in boilers during shut down.

c) It is required to warn of in-leakage of contaminants (primarily condenser cooling water ingress).

d) It helps us to develop the correlation with other chemistry parameters in case of aberration (i.e., cation conductivity).

e) To warn of condensate polisher malfunction.

f) To warn of make-up demineralizer malfunction.

6.8 CRUD, TOTAL IRON AND COPPER:

Monitoring CRUD, Iron, and Copper is necessary for the following reasons:

a) Provides an assessment of corrosion product transport in the cycle allowing for corrosion control optimization.

b) Iron and copper corrosion products form deposits on water wall tubes under high heat flux conditions, which can trap and concentrate impurities, contributing to tube failures by caustic gouging, acid phosphate corrosion, hydrogen damage, and other forms of localized corrosion. Iron and copper corrosion products also deposit on turbine surfaces.

c) Facilitates correlation between water chemistry parameters and plant operating variables, aiding in root cause analysis and performance optimization.

d) To check the accuracy of water chemistry control (such as reducing agent, oxygen, ammonia or pH), so ensuring that corrosion rates are kept at acceptable low levels. 5.10

6.9 TOTAL ORGANIC CARBON (TOC):

TOC monitoring is typically only required as a troubleshooting measure when other continuously monitored parameters are outside of target values. It is monitored because:

a) Organics may cause foaming in the boiler and high mechanical carryover, and may also affect the boiler and turbine corrosion, depending on the specific species or its thermal decomposition products that are present.

b) Organics can form organic acids and carbon dioxide from decomposition in the boiler and typically are detected by an elevation of cation conductivity.

c) To monitor make-up water quality for organic carbon content.

6.10 AMMONIA (NH_3) :

Monitoring of ammonia can be used to supplement the direct measurement of feedwater pH and/or specific conductivity for the control of the ammonia dosing rate. Ammonia monitoring is critical in mixed-metallurgy units to minimize corrosion of copper alloy components.

7.0 TYPES OF PARAMETERS IN CYCLE CHEMISTRY MONITORING

The chemistry parameters monitored under the cycle chemistry guidelines fall into two general categories:

i. CORE MONITORING PARAMETERS Monitoring these parameters is essential for optimum chemistry control. Core parameters represent the minimum level of surveillance required for all units and should be monitored continuously. PI tags must be available for all core parameters.

ii. DIAGNOSTIC MONITORING PARAMETERS These parameters are monitored to correlate water chemistry data and are essential for troubleshooting and identifying root causes of issues.

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

7.1 CORE MONITORING PARAMETERS The following table provides a comprehensive list of all core parameters along with their respective sample locations:

Table-1A: FOR DRUM TYPE UNITS (MIXED METALLURGY AND ALL-FERROUS UNITS)

Sl. No.	Parameters / Sample	Specific Conductivity	Cation Conductivity	Degassed cation conductivity	pH	Dissolved oxygen	Sodium	Phosphate	Silica	Chloride	ORP	Hydrazine
1	DM makeup	✓		✓								
2	CEP discharge		✓	✓		✓	✓					
3	SWAS-CPU*	✓	✓				✓					
4	D/A inlet										✓1	
5	D/A outlet					✓						
6	Feed water	✓	✓		✓	✓						✓1
7	Boiler water	✓	✓2		✓			✓	✓	✓		
8	Saturated steam		✓				✓					
9	Main steam		✓		✓		✓					
10	CPU Service vessel*	✓	✓				✓					
11	Hot-well (L/R)	✓										

*: For units equipped with CPU

✓1 : For mixed-metallurgy units

✓2: For units equipped with CPU

7.2 DIAGNOSTIC MONITORING PARAMETERS:

While customizing cycle chemistry programs, one or more diagnostic parameters should be monitored through offline testing in addition to the core parameters. The important diagnostic monitoring parameters are listed below:

i. Ammonia
ii. Iron and Copper
iii. CRUD
iv. Total Organic Carbon (TOC)
v. Low level Chloride

8.0 CHEMICAL CONTROL LIMITS OF STEAM-WATER CYCLE PARAMETERS:

8.1 CONDENSATE WATER

Table-2: Chemical Control Limits for Condensate Water Sample

SN	Parameter	Drum Boiler Units		Once-through Boiler Units			WHRB/ HRSG
		Mixed metallurgy	All-ferrous	Sub-critical	Super-critical		
		AVT-R	AVT-O	AVT-O	AVT-O	OT	AVT-O
1	pH, (25 ^{0}C)	9.0-9.3	9.2- 9.6	9.2-9.6	9.2-9.6 9.5-9.7*	8.5-9.0 9.5-9.7*	9.2-9.6
2	CC/DCC, µS/cm, (25 ^{0}C), max	0.30	0.20	0.20	0.15	0.15	0.30
3	DO, ppb, max	20	20	20	20	20	30
4	Silica (SiO_2), ppb, max	20	20	20	20	20	20
5	Sodium, ppb, max	2	2	2	2	2	2
6	Total Iron (dissolved & suspended), ppb, max	--	--	--	20*	20*	--
7	TOC, ppb, max	200	200	200	200	200	--

***:** For units with ACC (air-cooled condenser)

8.2 DEAERATOR INLET

Table-3: Chemical Control Limits for Deaerator Inlet Sample

Sl No	Parameter	Mixed Metallurgy Units
1	ORP at Deaerator inlet, mV (25 ^{0}C)	-250 to -150

8.3 FEED WATER

Table-4: Chemical Control Limits for Feed Water Sample

SN	Parameter	Drum Boiler Units		Once-through Boiler Units			WHRB/ HRSG
		Mixed metallurgy	All-ferrous	Sub-critical	Super-critical		
		AVT-R	AVT-O	AVT-O	AVT-O	OT	AVT-O
1	pH, (25 ^{0}C)	9.0-9.3	9.2- 9.6	9.2-9.6	9.2-9.6 9.5-9.7*	8.5-9.0 9.5-9.7*	9.2-9.6
2	Specific Conductivity, µS/cm, (25^{0}C)	3-5	5-11	5-11	5-11	2-3 9-14*	5-11
3	CC, µS/cm, (25 ^{0}C), max	0.30 0.20**	0.20	0.20	0.15	0.15	0.30
4	DO, ppb	< 5	5-10	5-10	5-10	50-150	5-10
5	Silica (SiO_2), ppb, max	20	20	20	20	20	20
6	Sodium, ppb, max	--	--	2	2	2	--
7	Total Iron (dissolved & suspended), ppb, max	10 (5**)	10 (5#)	5	5	2	20
8	Total Copper, ppb, max	5	--	-	--	--	--
9	Residual N_2H_4, ppb	10- 20	--	--	--	--	--

*: For units with ACC (air-cooled condenser)

**: For mixed-metallurgy units equipped with CPU (Ramagundam Stage-I)

#: For all-ferrous units equipped with CPU

Note:

i. Wherever a parameter limit is specified as a range, it is recommended to maintain the value close to the midpoint of that range. For example, if the limit is given as 9.2–9.6, it is preferable to maintain a value of 9.3–9.4 for AVT(O) operation, and around 8.8 for OT operation (for water-cooled condenser-based units only), with corresponding specific conductivities of approximately 5.0 µS/cm and 2.0 µS/cm, respectively.

ii. As feedwater can enter the steam path directly through attemperation sprays, feedwater sodium levels must be maintained within the recommended limits specified above. In the absence of a dedicated feedwater sodium analyzer, the sodium value obtained from the SWAS-CPU can be used as a reference for feedwater sodium. It must be ensured that the tapping point for the SWAS-CPU sample line is located downstream of the mixing header (at the point where the condensate and CPU common outlet headers merge).

iii. For drum units equipped with CPU:

Whenever the total iron content exceeds 5 ppb, both CPUs should be kept in service to maintain feedwater total iron below 5 ppb, in order to prevent iron deposition in the turbine through attemperation.

The feed water DO lower limit has been increased to 5 ppb for the AVT(O) regime to ensure the ORP always remains on the positive side.

8.4 BOILER WATER (FOR DRUM UNITS)

Table-5: Chemical Control Limits for Boiler Water Sample

SN	Parameters	For 110, 210, 250 MW Units		Kahalgaon Stg-I		For 490, 500, 600 MW Units		WHRB/HRSG		
				Boiler water	ODC water			LP drum*	IP drum**	HP Drum
		PT	CT	PT		PT	CT	PT	PT	PT
1	pH, (25 ^{0}C)	9.2-9.6	9.2-9.4	9.2-9.6	9.2-10.5	9.2-9.6	9.2 -9.4	9.2- 10	9.2-9.8	9.2- 9.6
2	Specific Conductivity, µS/cm, (25^{0}C)	6-20	6-15	5-10	5-50	6-15	6-10	60	15-25	40
3	CC, µS/cm, (25 ^{0}C), max	6	--	--	--	5	1.5	--	--	--
4	Chloride, ppb, max	500 200#	500 200#	150	500	200	200	1000	1000	1000
5	Phosphate, ppm	0.3-2.0	--	0.2-2.0	0.2-6.0	0.2-0.5	-	2.0-10.0	2.0-8.0	2.0- 6.0
6	Silica (SiO_2), ppb, max	To maintain Silica less than 20 ppb in steam								
7	Sodium, ppm	0.1-1.0	0.1-1.0	0.1-1.0	--	0.1-1.0	0.1-0.8	--	--	--

#: For Ramagundam Stg-I units

Note:

i. For units operating on Caustic Treatment (CT), the boiler pH should be maintained within the range of 9.2–9.4. In addition, sodium levels in saturated steam (SS) and main steam (MS) should be continuously monitored to detect any carryover and kept below 2 ppb.

ii. Recommended free sodium level (minimum 100ppb) should always be maintained in the boiler water to neutralize chloride and other anionic contaminants. However, its concentration must remain within the recommended range (as mentioned above) to prevent caustic gouging in boiler tubes and carryover into the steam path.

iii. During normal operation, phosphate shall be maintained at minimum level so that desired level of pH in boiler water is maintained to avoid any carryover. For any other abnormal conditions, such as condenser tube leakage etc., dosing of the boiler water with sodium hydroxide (ensure boiler sodium should be in the recommended limit as mentioned above) is also allowed as a supplement to TSP when needed for pH control.

iv. In phosphate treatment, the boiler water specific conductivity must always be greater than its cation conductivity. If Specific conductivity is less than Cation Conductivity, this indicates an upset condition (either contaminants such as chlorides and sulfates, or phosphate hideout return products). Immediate action is to be taken based on analysis to maintain the recommended parameters in boiler water.

v. Site to maintain data as per Annexure-I to capture and monitor phosphate hideout issues. This exercise should be carried out Once a week.

8.5 STEAM (SS, MS, & HRH)

Table-6: Chemical Control Limits for Steam Sample

SN	Parameters	Drum Boiler Units		Once-through Boiler Units			WHRB/ HRSG
		Mixed metallurgy	All-ferrous	Sub-critical	Super-critical		
		AVT-R	AVT-O	AVT-O	AVT-O	OT	AVT-O
1	pH, (25 ^{0}C)	9.0-9.3	9.2- 9.6	9.2-9.6	9.2-9.6 9.5-9.7*	8.5-9.0 9.5-9.7*	9.2-9.6
2	CC, µS/cm, (25 ^{0}C), max	0.30 0.20**	0.20	0.20	0.15	0.15	0.30
3	Silica (SiO_2), ppb, max	20	20	20	20	20	20
4	Sodium, ppb, max	2	2	2	2	2	2
5	Chloride, ppb, max	2	2	2	2	2	2
6	Total Iron (dissolved & suspended), ppb, max	5	5	5	5	2	--
7	Total Copper, ppb, max	5	--	-	--	--	--
8	TOC, ppb, max	100	100	100	100	100	--
9	CRUD, ppb	Nil					

*: For units with ACC (air-cooled condenser)
**: For mixed-metallurgy units equipped with CPU (Ramagundam Stage-I)

Note:

i. It is recommended to have a separate single channel online sodium analyzer for main steam sample for all drum units (490/500/600MW).

ii. If the steam (SS/MS/HRH) sodium level exceeds 2 ppb, immediate corrective action is required, as this indicates potential caustic carryover—a significant corrosion risk to the steam path. For drum-type units, boiler blowdown should be opened to maintain boiler water sodium concentration within recommended limit. For both drum and once-through type boiler units, the exhausted CPU (refer to Table 7) should be isolated immediately, and fresh CPU should be taken into service.

8.6 CPU SERVICE VESSEL

Table-7: Chemical Control Limits for CPU Service Vessel Sample

Parameters	**Drum units (PT/CT)**	**Once-through boiler units**	
		Sub-critical	Super-critical
Specific Conductivity, µS/cm, (25^{0}C)	2.0	0.50	0.10
CC, µS/cm (25 ^{0}C), max	0.10	0.10	0.10
Sodium, ppb, max	2	2	2
Chloride, ppb, max	2	2	2

NOTE:

i. For drum type units, at least one CPU service vessel at design flow is to be kept in service during normal operation.
ii. For once through units and sea water cooled condenser units, 100% condensate flow through CPU should be maintained during normal operation.

9.0 ACTION LEVELS:

There are four action levels which have been developed based on the following criteria for Main steam parameters:

i. Normal Values - Within Normal target values which are consistent with long term system reliability.

ii. Action Level 1 (AL-1)- (2 x Normal target value): Signifies that cycle chemistry is abnormal, or not in the target control range, and should be addressed by troubleshooting and resolving the issue to avoid increased risk of corrosion and/or deposition. For the critical core parameters like Cation conductivity, Sodium in Steam, if the parameter cannot be returned to normal within 48 hours the next action level requirements should be enacted (Action Level 2).

iii. Action Level 2 (AL-2)- (2 x Action level 1 value): Signifies that higher risk of corrosion and/or deposition exists and thus requires more aggressive corrective action measures to return the parameter(s) to AL-1 within 12 hours. For critical core parameters, if the parameter(s) cannot be returned to AL-1 within 12 hours, the next action level requirement should be enacted (Action Level 3) to escalate corrective actions to avoid increased risk of corrosion and/or deposition.

iv. Action Level 3 (AL-2)- (> Action Level 2 value): Signifies that active corrosion and/or deposition is occurring, and components are at high risk of damage. This action level indicates that the unit should be prepared for shutdown by reducing load and getting systems and equipment ready for shutdown unless the parameter(s) can be returned to AL-2 within 6 hours.

v. Action Level 4= Immediate shutdown-(>Action Level 3 value): For parameters with a designated Immediate Shutdown limit as specified in Table 9, if exceeded, the unit should be taken out of service immediately to dissipate heat from the boiler and stop steam flow to the turbine as quickly as possible, thereby preventing severe corrosion and/or deposition in the unit.

Table-8: Guideline Values for Deviation from Normal Operating Limits for Continuous Operation

Parameters (MAIN STEAM)	Unit	Normal Level	Action Level 1	Action level 2	Action Level 3	Action Level 4 = Immediate shutdown
Cation conductivity, (25°C)	µS/cm	≤0.15* ≤0.2**	≥ 0.2 < 0.35	≥ 0.35 < 0.5	≥ 0.5 < 1.0	≥ 1.0
Sodium (Na)	ppb	≤2	>2.0 ≤4.0	> 4.0 ≤8.0	>8	>20
Maximum exposure to contaminant conditions per event	h	--	≤ 48	≤ 12	≤ 6	0

* For Supercritical units
** For Subcritical units

Table-9: Immediate Shutdown Guidelines for All Types of Units

Unit Type	Immediate shutdown criteria
Drum boiler	If any one of the following conditions becomes true: **i.** Boiler water pH (measured at 25 ^{0}C) falls to 8.0 with further descending trend **ii.** Boiler water pH (measured at 25 ^{0}C) greater than 11.0 **iii.** When Main steam CC >1.0 µS/cm or Sodium >20 ppb
Once-through boiler (Subcritical and Supercritical)	If any one of the following conditions becomes true: **i.** Feed water pH (measured at 25 ^{0}C) falls to 7.0 with further descending trend. **ii.** Feed water or Main steam CC >1.0 µS/cm **iii.** Feed water or Main steam sodium >20 ppb ***Note:*** *For units where online feedwater sodium measurement is not available, the sodium value of the CPU outlet sample at SWAS shall be considered representative of the feedwater.*

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

10.0 RECORDS TO BE MAINTAINED

SN	Record description	Maintained by	Remarks
1	Continuous monitoring record of core parameters as mentioned in Section-6.1-Table-1A & 1B.	Chemistry/ C&I	
2	Availability of online analyzers as mentioned in Section-6.1-Table-1A & 1B.	Chemistry/ C&I	
3	Steam water cycle parameters of running units as per Sec- 7.0.	Chemistry	
4	Template for capturing phosphate hideout in drum boilers as specified in Annexure-I.	Chemistry	

11.0 Review

HOD (Chemistry) will review & take the approval of competent authority (HOP).

The document will be reviewed on 2 yearly basis or earlier, if required.

12.0 Checklist

Sl. No.	Checklist description	Reference clause in TC DOC: COS-ISO-00-OD/OPS/CHEM/ 001
1	Monitoring of core parameters and diagnostic parameters	Section-6.0
2	Availability of online analyzers for core parameters monitoring for all samples	Section -6.1
3	Maintenance of steam water cycle chemistry parameters limits	Section -7.0
4	Adherence to Action levels in case of parameter aberrations	Section -8.0
5	Records maintenance	Section -9.0

ANNEXURE-I: TEMPLATE FOR CAPTURING PHOSPHATE HIDEOUT IN DRUM BOILERS

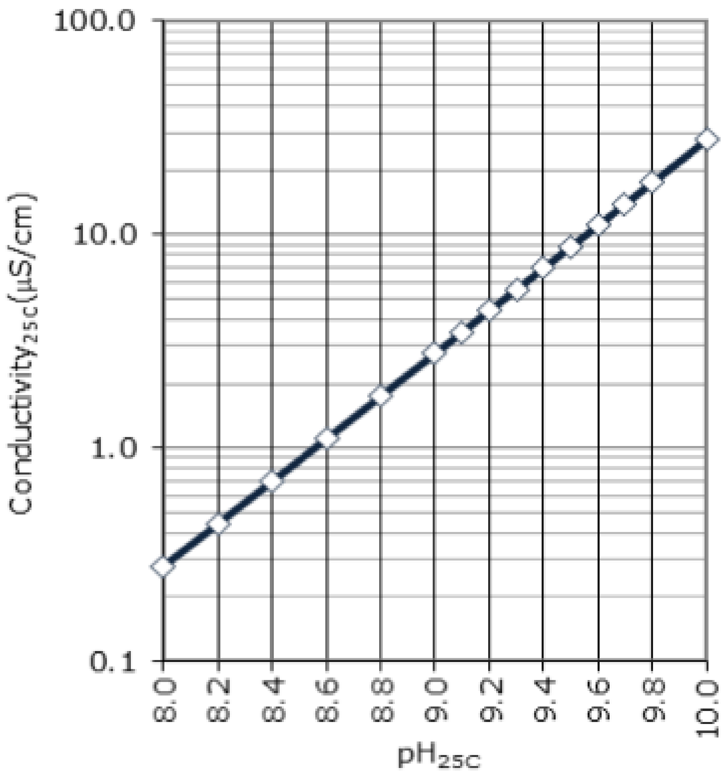
STATION NAME: UNIT:

Sl. No	Date/ time	Unit load	Drum pressure	Sample: Boiler Water				
				pH	SC, µs/cm (25 °C)	CC, µs/cm (25 °C)	Phosphate, ppm	Sodium (Na), ppb
1								
2								
3								

Note: Boiler phosphate hideout tests are to be carried out at different unit load preferably at full and technical minimum load.

ANNEXURE-II: SAMPLE WATER- AMMONIA-25 0C CONDUCTIVITY VERSUS pH.

The relationship is applicable under normal operating conditions but may not hold during startup due to CO_2 saturation. During normal operation, it remains valid for feedwater cation conductivity values up to 0.30 µS/cm.



pH	Conductivity (µS/cm)	Ammonia (ppm)
10.0	27.747	11.812
9.8	17.507	5.105
9.7	13.907	3.399
9.6	11.047	2.286
9.5	8.775	1.555
9.4	6.970	1.070
9.3	5.536	0.746
9.2	4.398	0.527
9.1	3.493	0.377
9.0	2.775	0.274
8.8	1.751	0.149
8.6	1.105	0.085
8.4	0.698	0.050
8.2	0.442	0.030
8.0	0.280	0.018

$$pH = 8.55 + \log(Cond_{25}) + 0.032(T-25)$$

Where: Cond is specific conductivity in µS/cm and T is temperature in °C)

TITLE	Chemical Control of the Water-Steam Circuits of Drum & Once Through Type Boilers and HRSG
LMI No.	LMI/OD/OPS/CHEM/001 Rev. No- 11

ANNEXURE-III: RECOMMENDED FEED WATER & BOILER WATER TREATMENT FOR DRUM UNITS.

Sl No.	Region	Station	Stage	Capacity (MW)	FEED WATER TREATMENT		BOILER WATER TREATMENT	
					Unit Startup and Normal Operation			
					AVT-R	AVT-O	PT	CT
1		Barauni	II	2x250		✓	✓	
2		Farakka	I	3x200	✓		✓	
3		Farakka	II	2x500		✓	✓	
4	ER-1	Farakka	III	1x500		✓	✓	
5		Kahalgaon	I	4x210	✓		✓	
6		Kahalgaon	II	3x500		✓	✓	
7		Kanti (MPTS)	II	2x195		✓	✓	
8	ER-2	Bongaigaon	I	3x250		✓	✓	
9		TSTPS	II	4x500		✓	✓	
10		Dadri Thermal	I	4x210		✓	✓	
11		Dadri Thermal	II	2x490		✓	✓	
12		Rihand	I	2x500		✓		✓
13		Rihand	II	2x500		✓	✓	
14		Rihand	II	2x500		✓	✓	
15		Singrauli	I	5x200	✓		✓	
16		Singrauli	II	2x500		✓	✓	
17	NR	Unchahar	I	2x210	✓		✓	
18		Unchahar	II	2x210		✓	✓	
19		Unchahar	III	1x210		✓	✓	
20		Unchahar	IV	1x500		✓	✓	
21		Vindhyachal	I	6x210	✓		✓	
22		Vindhyachal	II	2x500		✓	✓	
23		Vindhyachal	III	2x500		✓	✓	
24		Vindhyachal	IV	2x500		✓	✓	
25		Vindhyachal	V	1x500		✓	✓	
26		Ramagundam	I	3x200	✓		✓	
27		Ramagundam	II	3x500		✓	✓	
28	SR	Ramagundam	III	1x500		✓	✓	
29		Simhadri	I	2x500		✓	✓	
30		Simhadri	II	2x500		✓	✓	
31	WR-1	Mouda	I	2x500		✓	✓	
32		Korba	I	3x200	✓		✓	
33	WR-2	Korba	II	3x500		✓	✓	
34		Korba	III	1x500		✓	✓	
35		Sipat	II	2x500		✓	✓	
36		BRBCL	I	4x250		✓	✓	
37	JV	Jhajjar	I	3x500		✓	✓	
38		Vallur	I	3x500		✓	✓	